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Risky choice and diminishing sensitivity in MaaS context: A nonlinear logit analysis of traveler behavior

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ABSTRACT

In the rapidly transforming landscape of urban transportation, Mobility-as-a-Service (MaaS) has emerged as a potentially transformative solution to adapt to fluctuating travel demand. However, despite arguments in the literature suggesting that uncertainty in travel plans may influence travelers' preferences for MaaS, the extent and mechanisms of this impact remain relatively underexplored. This study aims to bridge this gap by evaluating travelers' preferences toward MaaS subscriptions, mode choices, and their reactions to uncertainty within a risky-choice modeling framework. We incorporate a linear-specified benchmark model and two nonlinear models (one based on Expected Utility Theory with Constant Absolute Risk Aversion adjustments, and the other on Prospect Theory), to estimate travelers' risk perceptions and their valuation of gains and losses associated with travel time and costs caused by occasional activities during commute trips. The data utilized in our study originates from a specifically designed stated preference survey carried out in Hong Kong. Our findings present insightful observations into MaaS bundle configurations, socio-demographic patterns, and notably, behaviors under uncertain circumstances. In the face of uncertainty, we observe a noticeable increase in reluctance to adopt MaaS, and a heightened preference for taxis. The models also underscore significant risk aversion and diminishing sensitivity concerning travel time and cost, with these tendencies proving more pronounced for travel time compared to cost. These insights are valuable for policymakers and MaaS operators, offering an enhanced understanding of users' risk attitudes and the intricate interplay between travelers' preferences for MaaS and the uncertainties inherent in daily travel.

1. Introduction

Uncertainty invariably introduces complexity into the decision-making process. In the classical transportation field, extensive research has focused on behavioral reactions to travel time reliability, primarily dealing with supply-side uncertainties such as unexpected disruptions causing road capacity reduction (see Carrion & Levinson (2012) for a review). However, the demand side of travel is equally affected by uncertainty, a consequence of the activity-based nature of travel demand that frequently involves unpredictable

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or unplanned activities (Bhat and Koppelman, 2003). This unpredictability introduces significant variability and uncertainty into travel behavior (Ma et al., 2021), influencing both immediate transport mode choices and long-term mobility resource ownership (Le Vine et al., 2013).

The emergence of Mobility-as-a-Service (MaaS) offers a novel context for accommodating this uncertainty. MaaS typically refers to a digital-platform-based service that allows users to plan, book, and pay for a variety of mobility services, often offering subscription-based discounts (Esztergár-Kiss and Kerényi, 2020; Smith and Hensher, 2020). MaaS provides travelers with a diverse mobility portfolio, including bundling options that make flexible transport modes (e.g., taxis) more affordable, thereby enriching choice diversity. While research suggests that the preference for MaaS bundles is influenced by uncertainty (Hensher et al., 2021b), there is only a limited amount of research that thoroughly quantifies the impact of uncertainty in travel plans on travelers' subscription preferences and subsequent mode choices (Liu et al., 2023). This research gap primarily stems from data scarcity and the complexity involved in defining, measuring, and evaluating uncertainty, particularly when considering multiple trips over a subscription period in the MaaS context.

Occasional activities (such as detour shopping, social gatherings, and business meetings), which are unforeseeable at the time of deciding whether to subscribe, are one of the major sources of uncertainties within the MaaS context (Liu et al., 2023). This uncertainty manifests in both short-term and long-term scenarios. In the short term, these activities can lead to increased travel costs and time, potentially resulting in delays. This impact is particularly acute during commute trips, where there is a heightened sensitivity towards transport unreliability (Schwanen and Ettema, 2009). Over the long term, the frequency of these activities complicates decision-making for commuters, who must balance the benefits of MaaS subscriptions (such as enhanced affordability of flexible transport options to counter unreliability) against potential losses (such as unused MaaS credits due to overestimated travel demand at the time of subscription).

To address this research gap, we designed a stated preference (SP) survey to gather insights into the impact of MaaS bundles on travelers' mode choices in the context of occasional activities over a more extended period (a week). In our previous study (Liu et al., 2023), we calibrated the travelers' risk attitudes using choice information from lottery games. However, individuals may exhibit different risk attitudes in lottery games and MaaS bundle choice tasks. Furthermore, the previous models restricted our ability to examine different levels of uncertainty within a unified framework. In response to these challenges, this study includes a benchmark model grounded in Expected Utility Theory (EUT) with a linear specification, alongside two advanced models combining nonlinear transformations: one based on EUT with adjustments through Constant Absolute Risk Aversion (CARA), and the other based on Prospect Theory. First, we assess how uncertainties cause deviations in consumers' preferences for MaaS subscriptions and transport modes. Second, we explore travelers' risk attitudes toward uncertainties in travel time and costs in upcoming MaaS subscription periods. Lastly, we shed light on how travelers perceive gains and losses related to travel time and expenses.

The remainder of this paper is organized as follows: section 2 offers a comprehensive review of existing literature, focusing on MaaS demand modeling and decision-making analytical frameworks under uncertainties. Section 3 details our survey design, describes the collected data, and presents preliminary analysis findings. Our modeling frameworks are detailed in Section 4, followed by a discussion of modeling results in Section 5. Section 6 explores policy implications, study limitations, and outlines potential future research directions. The paper concludes in Section 7 by summarizing the key findings.

2. Literature review

2.1. MaaS demand modeling

Ever since Heikkilä (2014) first introduced the concept of MaaS, there has been a significant increase in research centered on this promising paradigm. This surge in interest spans both supply (Ding et al., 2023b; Mulley et al., 2020; Musolino et al., 2022), and demand sides, especially in evaluating potential market size and user engagement (Alyavina et al., 2020; Karlsson et al., 2020; Polydoropoulou et al., 2020). While a substantial body of this literature employs discrete choice models to examine MaaS demand, data collection methods employed vary widely, spanning interviews (Alyavina et al., 2020; Schikofsky et al., 2020), Likert-scale surveys (Alonso-González et al., 2020), focus groups (Fioreze et al., 2019), stated adaptation experiment (Feneri et al., 2022), and revealed preference surveys (Guidon et al., 2020; Hasselwander et al., 2022; Karlsson et al., 2016). Among these data collection methodologies, SP surveys have emerged as the most prevalent (Caiati et al., 2020; Ho et al., 2018; Matyas and Kamargianni, 2021; Mulley et al., 2020; Tsouros et al., 2021). The popularity of SP surveys can be attributed to their distinct advantages including the capacity to elicit insights into hypothetical scenarios that are not readily observable in real-world settings, and the ability to quantify the trade-offs individuals are willing to make when faced with different choice alternatives. These attributes are particularly beneficial considering the limited real-world implementation of MaaS, thereby offering researchers a valuable instrument for comprehending the dynamics of MaaS demand and user preferences.

Travelers' tendency toward MaaS subscription exhibits variations within the literature. Some studies reflect a generally positive inclination toward MaaS adoption (Alyavina et al., 2020; Hensher et al., 2021a; Karlsson et al., 2016), whereas Fioreze et al. (2019) report contrasting findings. Regarding potential uptake, the majority of studies report values ranging between 40 % and 60 % (Alonso-González et al., 2020; Ho et al., 2018, 2020; Matyas and Kamargianni, 2021). However, Caiati et al. (2020) report that only around 17 % of respondents expressed interest in MaaS subscriptions. Such divergences could be attributed to variations in the study area and data collection method. Among the various factors affecting travelers' propensity to subscribe to MaaS, socio-demographic characteristics consistently emerge as one of the most influential factors (Hasselwander et al., 2022; Kim et al., 2021; Schikofsky et al., 2020; Wang et al., 2022). Yet, the effects of these socio-demographic factors vary, as Kriswardhana and Esztergár-Kiss (2023) revealed

through their review of 29 articles. Matyas and Kamargianni (2021), through a latent class choice model, have shown that socio-demographic features can be utilized to categorize individuals' attitudes toward MaaS bundles into three groups: MaaS avoiders, MaaS explorers, and MaaS enthusiasts.

Researchers have observed a substantial shift in the preferred transport mode upon subscribing to MaaS. Karlsson et al. (2016) highlight an increased utilization of alternative modes over private cars, drawing from evidence in the UbiGo trial. This observation aligns with the findings of Liu et al. (2023), who noted a distinct preference for taxis over private vehicle usage subsequent to MaaS subscription. In contrast, Hensher et al., (2021a) report an inclination towards increased private car use after MaaS subscription in the Sydney MaaS trial. Additionally, Matyas and Kamargianni (2019) reveal that respondents exhibited an inclination to explore modes of transportation that they had rarely used before, if these modes were incorporated into the MaaS plan.

2.2. Choice under Uncertainty: EUT and prospect theory

Expected Utility Theory (EUT), which posits that individuals rationally select the option with the highest expected utility, has found extensive application in fields as diverse as psychology (Caplin and Leahy, 2001; Kate Bundorf et al., 2013), sociology (Becker, 2003; Harrison and Rutström, 2009), and economics (Kubler et al., 2014; Qazi et al., 2018) since its axiomatization by Von Neumann and Morgenstern (1944). The tractability and intuitive appeal of EUT grant it distinct advantages. In transportation research, EUT is often integrated with random utility models to handle discrete choice problems, further extending to encompass non-linear transformation for the estimation of risk attitudes (averse, neutral, or seeking) under uncertainty (Liu and Polak, 2007). This general form of EUT has seen application in areas like car-sharing vehicle allocation (Wu et al., 2019, 2022), dynamic pricing of electric vehicles (Latinopoulos et al., 2017), matching for ride-sourcing platforms (Ding et al., 2023a), and risk attitudes calibration (Dixit et al., 2019a; Dixit et al., 2015; Wijayaratna and Dixit, 2016).

However, human decision-making does not always adhere to complete rationality. Cognitive load may lead us to either overlook or over-utilize some information or settle for satisfactory outcomes based on initial intuition (Aronson, 2007). Consequently, despite the merits of EUT, empirical studies have identified several violations of its foundational axioms, as seen in the well-known examples of the Ellsberg paradox[1] and the Allais paradox[2]. In response to these shortcomings, alternative analytical frameworks have been developed, such as rank-dependent expected utility (Quiggin, 2012) and regret theory (Loomes and Sugden, 1982). Among these, cumulative prospect theory has gained notable attention for its ability to explain these paradoxes (De Palma et al., 2008), positing that utility maximization occurs after two cognitive phases: editing and evaluating the decision problem (Tversky and Kahneman, 1992).

Cumulative prospect theory has gradually garnered recognition in transportation research. Avineri and Prashker (2005) compared experimental results of route-choice decision-making with cumulative prospect theory predictions. They found that as the variance in travel times increases, travelers' sensitivity to differences in travel time diminishes. Schwanen and Ettema (2009) investigated parents' responses to unreliable transportation when chauffeuring children, underscoring the applicability of cumulative prospect theory in depicting activity-travel behavior under uncertainty. Similarly, Latinopoulos et al. (2017) compared the estimation results of prospect theory with other commonly used models (EUT and rank-dependent EUT) in the context of dynamic pricing for parking and charging of electric vehicles. Rose and Masiero (2010) accessed several theoretical developments of prospect theory (i.e., reference dependence, loss aversion, and diminishing sensitivities) in transportation policy evaluation, with a specific focus on willingness-to-pay and willingness-to-accept rather than on marginal utilities.

The rapidly growing field of MaaS research has yielded a wealth of insights regarding demand modeling. However, a notable gap in this body of research lies in the consideration of uncertainty, which can have a profound impact on MaaS preferences. For instance, ambiguity surrounding future travel plans might critically impact an individual's decision to subscribe to MaaS. While the role of uncertainty in decision-making processes has been acknowledged by researchers such as Hensher et al. (2021b), there remains a scarcity of rigorous exploration in this particular aspect.

3. Data

The data used in this study is obtained from a SP survey conducted in Hong Kong. Details of the survey can be found in Liu et al. (2023). After introducing MaaS as "a digital platform that enables access to common transport modes through a single app, offering travelers the option to purchase pre-paid MaaS bundles for discounted access", we provide further clarification by offering an example to help respondents better understand how MaaS works (please refer to Appendix A). The survey proceeded to collect four main parts of information from the respondents, including (1) socio-demographic features such as age, gender, education level, and income; (2) current commute travel information, such as the distribution of weekly transport modes, the total number of commute trips per week, and commute duration; (3) stated choice tasks regarding MaaS subscription with uncertain travel time and cost; and (4) nine lottery games where participants had to choose between two lotteries (Dixit et al., 2019b). The third part is the core of the survey, in which we

[1] When faced with a choice between a gamble with known probabilities and one with uncertain probabilities, individuals tend to choose the former, indicating a preference for certainty and an aversion to situations with ambiguous or uncertain probabilities. Please refer to Ellsberg (1961) for more details.

[2] Individuals may exhibit context-dependent decision-making when faced with choices involving risks and uncertainties. In one situation, they might opt for the certain outcome even if it has a lower value. However, in a similar context, they could prefer the riskier option. Please refer to Allais (1953) for more details.

created an SP experiment using D-efficient designs (Rose et al., 2009). Initial priors were derived from the existing literature (Ho et al., 2018; Wu et al., 2019) and subsequently refined using regression results estimated from preliminary pilot data.[3] The attributes and the levels we selected to describe the MaaS bundles are presented in Table 1.

We categorized respondents into three distinct groups to ensure the MaaS plans proposed were realistic. These groups included: the "short-trip block" for those with an estimated weekly total commute time of less than 300 min, the "medium-trip block" for commute time between 300 and 600 min, and the "long-trip block" for individuals with more than 600 min of estimated weekly commute time. We exercised careful consideration in designing the attribute levels. Considering that previous studies often provide free miles or hours, akin to a 100% discount with usage limitations (Caiati et al., 2020; Ho et al., 2018), we set the maximum discount for taxi fares at 80%. This aims to attract customers while preventing potential market disruption and environmental unsustainability resulting from overly generous taxi discounts. We intentionally structured these fare discounts to slightly favor PT over taxis, aiming to promote sustainable mobility. Furthermore, we aimed to quantify the potential of certain MaaS features—such as advance booking, seamless transfer, and prioritized matching—to reduce travel time, by meticulously designing the time-saving factors. Recognizing the inherent flexibility of taxis, we assigned a slightly higher time-saving factor to them compared to PT, with the upper limit set at 35% (a 20-minute taxi journey can be reduced by 7 min, which is attainable through prioritized taxi matching for MaaS users). The "perceived probability" is reported by respondents based on their real experience of encountering occasional activities in every ten typical commute trips (the mean value is 22.7% in our data).

Each participant encountered six choice tasks, and an example of these tasks is depicted in Fig. 1. At the top of Fig. 1, respondents are presented with the probability of their reported occasional activities, accompanied by the estimated weekly cost and per-trip travel time of their current travel patterns. Both cost and time are represented as triangular distributions with lower bounds, upper bounds, and most likely values[4], reflecting the uncertainties associated with occasional trips. Subsequently, a MaaS bundle is presented to the respondent, and they can adjust their mode usage by manipulating the sliders (all initially set to 0).

The transport mode choices in the survey encompassed six prevalent modes in Hong Kong: driving, taxi, carpool, bus (solely), rail (solely), and a combined use of bus and rail. These modes, including PT, are widely accessible throughout Hong Kong. The city's PT system, in particular, is renowned for having one of the highest coverage rates globally (Bloomberg, 2022). However, it is worth noting that typical carpool service providers are not popular in Hong Kong due to their illegality without a hire-car permit. Consequently, for the purposes of this survey, carpool refers specifically to the Company's Shuttle Bus and Residents' Bus services. These services share similarities with broader carpool interpretation[5], such as shared car journeys with minimal or no intermediate stops. The "taxi" category encompasses the relatively small portion of trips that are conducted using non-shared ridesourcing services. Furthermore, the combination of bus and rail was included as an option to mirror respondents' actual travel experiences, especially in scenarios where solely using one mode might require extensive walking. We clearly defined these transport modes in the survey to ensure respondents had an accurate understanding. Respondents were asked to choose based on the transport modes they typically use for commute.

We assume that the number of weekly commute trips for each respondent does not vary before and after the introduction of the MaaS bundle, so the aggregate of their selected frequencies across all modes should match the total weekly trip count they initially reported in the survey (i.e., 10 times in Fig. 1). For ease of comparison, the travel times and costs are updated in real time, allowing respondents to assess the differences in travel time and cost before and after purchasing MaaS and adjusting mode usage (as illustrated by the 257 HKD and 29 mins beneath the sliders in Fig. 1). At the bottom of Fig. 1, respondents can indicate whether to subscribe to the MaaS bundle or not.

The survey was implemented using the online platform *Qualtrics*. Respondents who finished the survey within excessively short response times (under 5 min) were deemed "unserious", and their responses were removed in the subsequent data analyses. In July 2021, a sample of 1,007 valid respondents was recruited in Hong Kong by a marketing research firm, Consumer Search Group (CSG). After eliminating ten incomplete or irrational responses[6], the final sample size was 997. Our focus was on regular commuters. Consequently, all participants were either employed adults (including full-time, part-time, or self-employed) or university students. Table 2 presents an overview of the socio-demographic features of our sample and the general population of Hong Kong. As our sample does not include children, the education level and driving license ownership of our sample are thus higher than the overall Hong Kong population. Aside from these aspects, our sample is representative of the general Hong Kong population.

4. Methodology

We have meticulously evaluated the selection of models. Firstly, models based on EUT are indisputably prevalent in the related literature. Nonetheless, the inclusion of a non-EUT-based model is imperative, as it can encompass certain dimensions that EUT

[3] The priors employed are as follows: travel time, − 0.789; travel cost, − 0.755; bundle price, − 0.519; and the constant for MaaS bundle, − 0.152.

[4] The lower bound is calculated assuming that none of the trips involve occasional activities. The upper bound is derived under the assumption that all trips include occasional activities. The most likely value is determined based on the occurrence of occasional activities strictly at the assumed probability (e.g., at the top of Fig. 1, calculated as $0.5 \times 434 + 0.5 \times 707 = 571$).

[5] The term "carpool" varies subtly in definition across academic articles, and in certain contexts, these services also share characteristics with "microtransit" (Shaheen & Cohen, 2019).

[6] We excluded observations from one respondent due to illogical responses (the individual reported having two household members but three children). Additionally, data from nine respondents were omitted because they were incomplete, specifically lacking information about occasional activities.

Table 1
Attributes and their levels.

Attribute	Block	Levels
Bundle Price	Short-trip	$50, $100, $150
Bundle Price	Medium-trip	$100, $150, $200
Bundle Price	Long-trip	$200, $250, $300
Fare-discount for taxi	All	20% off, 50% off, 80% off
Fare-discount for PT and carpool	All	25% off, 50% off, 100% off
Time-saving for taxi	All	15%, 25%, 35%
Time-saving for PT and carpool	All	10%, 15%, 20%
Occasional activity probability	All	0%, 50%, perceived probability

models, constrained by their inherent assumptions, are unable to capture. Employing these two distinct types of models together can enhance the robustness of our analysis.

Within the non-EUT-based models, while those based on regret theory allow for the observation of travelers' regret over unchosen alternatives under uncertainty, it may not be entirely applicable to our specific context. Specifically, one of the core assumptions of regret theory is the ability of individuals to evaluate possible outcomes of different choices (Chorus et al., 2008; Herweg and Müller, 2021; Kim et al., 2017). In our scenario, although there appear to be only two alternatives (subscribing or not subscribing to MaaS), the outcomes of each alternative (travel time and costs) are intricately linked to how participants use various transport modes during the week, resulting in a wide range of possible values[7]. In contrast, models based on prospect theory require comparison with a reference point. In this study, respondents would naturally compare their benefits or losses after subscribing to MaaS against the *status quo*, making it intuitively the reference point.

This section briefly outlines the three models we constructed for this study. We first introduced an EUT-based benchmark model with a linear utility function in Section 4.1. Building upon this benchmark model, we advanced two more models incorporating non-linear transformations of both travel time and travel cost. Section 4.2 elaborates on the EUT-based CARA model, while Section 4.3 details the Prospect Theory-based model. To accommodate heterogeneity, random parameters have been incorporated into each of the three models.

4.1. EUT-based benchmark model

Occasional activities during routine commutes inevitably lead to increased travel time and cost. The resulting uncertainty, originating from such events, impacts individuals' sensitivity to travel time and cost, as well as their preferences for MaaS subscription and assorted transport modes. Depending on the probability θ of occasional activity a ($a = 0$: no occasional activity; $a = 1$: occasional activity occurs with probability), the utility of MaaS subscription that a traveler obtains from an alternative j ($j = 0$: pay-as-you-go; $j = 1$: subscribe to a MaaS bundle) could be simply derived from EUT as:

$$U_j = \beta_t(\theta T_{a=1} + (1-\theta)T_{a=0}) + \beta_c(\theta C_{a=1} + (1-\theta)C_{a=0}) + \boldsymbol{\beta_1 Z} + \boldsymbol{\beta_2 P} + \boldsymbol{\beta_3 S} + \varepsilon_{ja} \tag{1}$$

where T and C are weekly travel time and cost, respectively. The travel time and travel cost in scenarios with ($a = 1$) and without ($a = 0$) occasional activities are different due to the potential need for detours. $\boldsymbol{\beta}_1$ is the vector of coefficients corresponding to vector $\boldsymbol{Z}$ that represents MaaS bundle attributes (including an alternative-specific constant term denoting consumer preference for subscription, *ceteris paribus*), and $\boldsymbol{\beta}_3$ is the vector of coefficients corresponding to the socio-demographic vector $\boldsymbol{S}$. $\boldsymbol{\beta}_2$ signifies a coefficient vector that embodies the preference for various transport modes. These modes, represented by vector $\boldsymbol{P}$, are expressed by their proportionate weekly usage. ε_{ja} is the error term following Gumbel distribution and assumed to be identically and independently Gumbel distributed (IID) over alternatives.

To capture changes in travelers' preference for MaaS subscription and transport mode choices, we also introduce "deviation" variables (denoted as $\beta_{deviation}$). These variables apply to the alternative-specific constant (ASC) for MaaS subscription from $\boldsymbol{\beta}_1$ and the coefficients for mode preference from $\boldsymbol{\beta}_2$. That is, for each coefficient β mentioned above:

$$\beta = \begin{cases} \beta_{base}, & a = 0 \\ \beta_{base} + \beta_{deviation}, & a = 1 \end{cases} \tag{2}$$

For variables of particular interest (e.g., travel time and travel cost, the "deviation" variables mentioned in Eq. (2), and the nonlinear transformation parameters that will be introduced in Sections 4.2 and 4.3), we addressed preference heterogeneity using random parameters:

$$\boldsymbol{\beta} = \overline{\boldsymbol{\beta}} + \boldsymbol{\sigma v}_i \tag{3}$$

[7] For example, assume that an average respondent commutes 10 times per week, and considering the need to distribute these 10 trips across the 6 available transport modes, there will be 3,003 possible permutations.

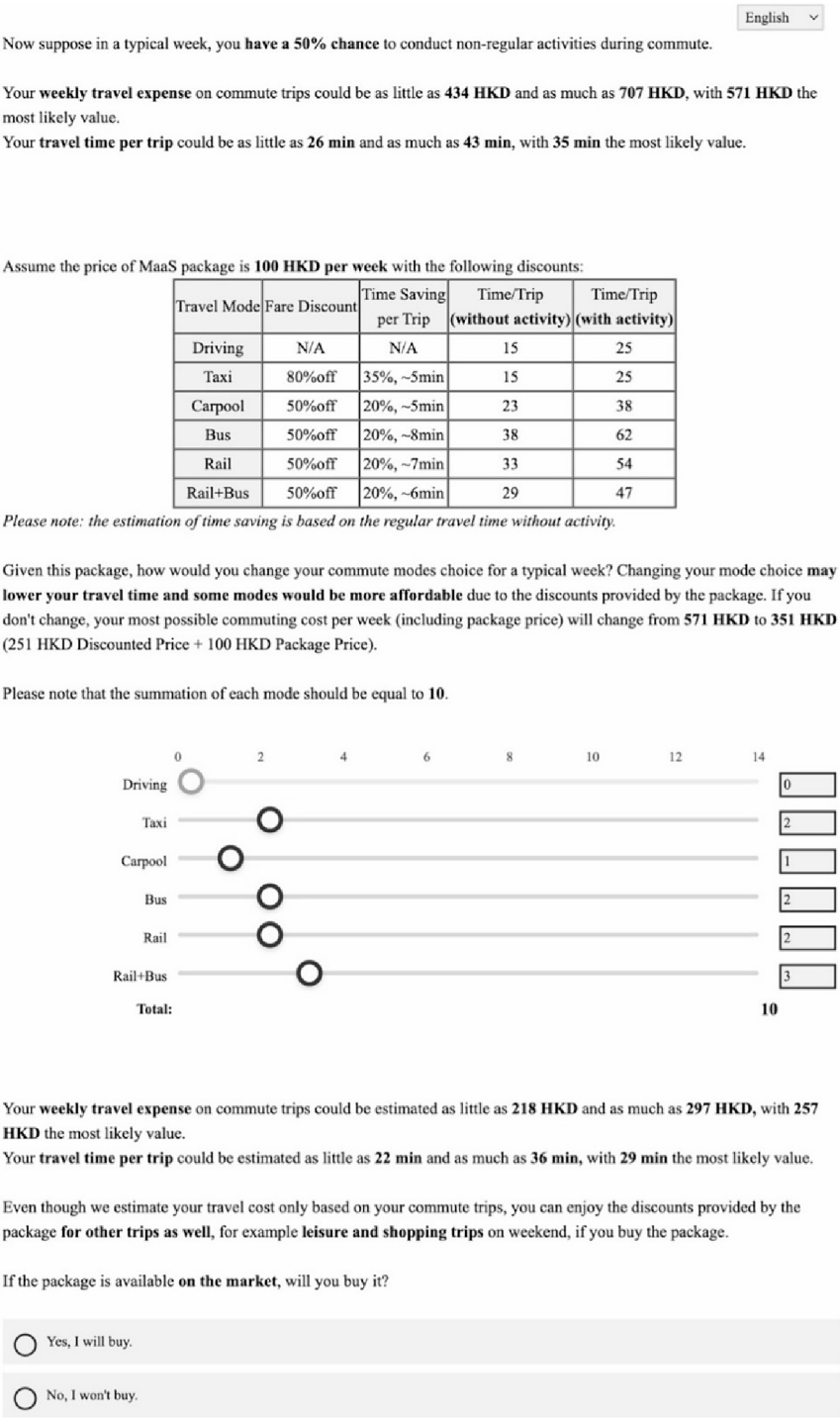


Fig. 1. An example of the stated preference choice tasks.

where the vector $\overline{\boldsymbol{\beta}}$ represents the population mean values, $\boldsymbol{\sigma}$ denotes the vector of standard deviations around these mean values, and $\boldsymbol{v_i}$ embodies the individual-specific heterogeneity, characterized by a mean of zero and a standard deviation of one.

Conditional on $\boldsymbol{\beta}$, the probability of an individual choosing alternative j follows the standard logit formula:

$$L_j = \frac{e^{\boldsymbol{\beta X}_j}}{\sum_{i=1}^{I} e^{\boldsymbol{\beta X}_i}} \tag{4}$$

Table 2
Descriptive statistics of the sample and Hong Kong population.

	Our sample	Hong Kong population (2021)
Male	45.4%	45.6%
Female	54.6%	54.4%
Below 35 years old	34.6%	34.2%
Above 35 years old	65.4%	65.8%
Household car ownership	30.5%	N/A
Driving licenses ownership	42.3%	30.9%
Education: below high school	2.0%	32.3%
Education: high school	35.5%	33.2%
Education: Bachelor	50.8%	34.5%
Education: Master or above	11.7%	
Household size: ≥ 6	2.4%	3.2%
Household size: < 6 (mean)	3.1	2.7
The average number of children (age < 18) in household	0.4	0.4

Note: Hong Kong statistics were retrieved from the Census and Statistics Department (2021) and the Transport Department (2021).

where $\boldsymbol{X}$ represents all variables considered in Eq. (1). I is the total number of alternatives ($I = 2$ in this case). Assuming $\boldsymbol{\beta}$ follows the distribution $f(\boldsymbol{\beta}|\boldsymbol{\theta})$, where $\boldsymbol{\theta}$ represents the parameters of the distribution of $\boldsymbol{\beta}$ over the population, the probability of an individual choosing alternative j is the integral of the above logit formula over the probability distribution of $\boldsymbol{\beta}$:

$$P_j = \int L_j(\boldsymbol{\beta}) f(\boldsymbol{\beta}|\boldsymbol{\theta}) d\boldsymbol{\beta} \tag{5}$$

4.2. EUT-based CARA model

To capture user's risk preferences, we further introduce Constant Absolute Risk Aversion (CARA) non-linear transformation into Eq. (1). Earlier studies have adopted this approach to estimate the impact of a single dimension of uncertainty (Hensher et al., 2011) or dual dimensions of uncertainties (Wu et al., 2019). Assuming travelers have a non-identical risk attitude towards travel time and cost:

$$U(T_a) = \left(1 - e^{-r_t * T_a}\right)/r_t \tag{6}$$

$$U(C_a) = \left(1 - e^{-r_c * C_a}\right)/r_c \tag{7}$$

where r_t and r_c represent the degrees of risk preference for travel time and cost, respectively. A value of $r > 0$ indicates risk aversion, while $r < 0$ signifies risk seeking. The $U(T_a)$ and $U(C_a)$ terms substitute the T_a and C_a terms in Eq. (1), respectively.

4.3. Prospect theory-based model

Prospect theory seeks to reconcile the contradictions of EUT by adjusting the weighting function and the value function (the evaluation phase) after simplifying the decision problem with respect to a reference point (the editing phase) (Kahneman and Tversky, 1979). In the conventional prospect theory model, the weighting function should permit the conversion of probabilities into weights, to compensate for the overestimation of small probabilities and the underestimation of large probabilities. However, in our study, we only consider three types of probabilities: 0 %, 50 %, and the perceived probability reported by the respondents. Given Tversky and Kahneman (1992)'s postulation that a 0 % probability should be precisely transformed to 0 %, and a 50 % probability should not deviate significantly from 50 %, and bearing in mind that the perceived probabilities ought to mirror the respondents' actual weights, we opt against transforming probabilities via a weighting function in our study to prevent excessive complication of the problem. A comparable decision is evident in Latinopoulos et al. (2017).

The value function assumes a two-part utility function:

$$v(x) = \begin{cases} x^{\delta}, if x \geq 0 \\ -\lambda(-x)^{\zeta}, if x < 0 \end{cases} \tag{8}$$

where x is the relative outcome gain/loss with respect to the reference point, λ is the parameter for loss aversion, δ and ζ are estimable coefficients reflecting the diminishing sensitivity. Moreover, we assume $\delta = \zeta$[8] as some previous work illustrated (Schwanen and Ettema, 2009; Tversky and Kahneman, 1992), while setting different values of δ for travel time and cost like the EUT models (see Eq. (6) and Eq. (7)). The utility function is now:

[8] We adopted this approach after evaluating a model premised on $\delta \neq \zeta$. In such a scenario, the dual dimensions of uncertainty in our study would lead to four parameters symbolizing diminishing sensitivity (two for time, and two for cost), thereby overly complicating the model and hindering convergence.

$$U_j = \theta\left[\beta_{c(gain)}(CD_{a=1}i_1)^{\delta_c} - \beta_{c(loss)}(-CD_{a=1}i_2)^{\delta_c} + \beta_{t(gain)}(TD_{a=1}i_3)^{\delta_t} - \beta_{t(loss)}(-TD_{a=1}i_4)^{\delta_t}\right] + (1-\theta)\left[\beta_{c(gain)}(CD_{a=0}i_5)^{\delta_c} - \beta_{c(loss)}(-CD_{a=0}i_6)^{\delta_c} + \beta_{t(gain)}(TD_{a=0}i_7)^{\delta_t} - \beta_{t(loss)}(-TD_{a=0}i_8)^{\delta_t}\right] + \boldsymbol{\beta}_1\boldsymbol{Z} + \boldsymbol{\beta}_2\boldsymbol{P} + \boldsymbol{\beta}_3\boldsymbol{S} + \varepsilon_{ja} \quad (9)$$

where *CD* and *TD* are weekly travel costs and time differences compared to the *status quo*, respectively. i represents a set of binary indicators organized into distinct pairs like i_1 and i_2, i_3 and i_4, continuing through to i_7 and i_8. Within each pair, both indicators cannot simultaneously be 1. For example, $i_1 = 1$ only if $CD_{a=1} > 0$, while $i_2 = 1$ only if $-CD_{a=1} > 0$. This logic is consistently applied across all pairs.

Similar to the EUT-based model, we include the preference deviation under uncertainty ($\beta_{deviation}$) as specified in Eq. (2).

As presented in Section 4.1, we used mixed logit models to construct choice models based on both the EUT-based CARA utility and the prospect theory-based utility functions. The models were calibrated using Biogeme 3.2.12 (Bierlaire, 2020), employing 3,000 draws for Monte-Carlo integration, and assuming that all random variables follow normal distributions.

5. Results

The willingness to purchase the MaaS bundle among respondents was 58.6 % overall. In the short-trip (28.4 % of respondents), medium-trip (48.8 %), and long-trip (22.8 %) blocks, the willingness rates were 52.6 %, 60.0 %, and 63.1 %, respectively. This section then presents the analyses of the results estimated from the three models introduced in Section 4, with Table 3 showing their estimated outcomes. The level of confidence in significance is indicated by *** ($p < 0.01$), ** ($0.01 \leq p < 0.05$), and * ($0.05 \leq p < 0.1$). The variables incorporated into these models are categorized into four groups: MaaS bundle attributes, socio-demographic characteristics, mode-preference constants, and non-linear transformation parameters. We present results relevant to variables that are not influenced by uncertainties in Section 5.1, while those that are impacted by uncertainties are presented in Section 5.2.

In general, the benchmark model exhibits relatively low goodness-of-fit, while the prospect theory-based model shows the best goodness-of-fit in terms of adjusted Pseudo-r^2. Given that these models yield nearly identical results in terms of parameter signs and statistical significance, we jointly present their results in the subsequent subsections.

5.1. Variables unaffected by uncertainties

Aligning with the findings of Ho et al. (2018) and Caiati et al. (2020), we observe the ASC for MaaS subscription to be negative and statistically significant. This suggests that all else being equal, travelers tend to prefer the conservative alternative (pay-as-you-go in this study). The noteworthy negative coefficient for the bundle price, coupled with the positive coefficients for fare discounts, intuitively indicate that respondents prefer cost-effective bundles with higher fare discounts. However, when considering the time-saving feature of MaaS, respondents primarily focus on the potential time savings from taxi services, while somewhat displaying disutility towards time savings from PT or carpool. This can be rationalized by the fact that PT riders are generally less time-sensitive compared to taxi riders. Furthermore, if travelers wish to optimize the efficiency of their journey, they are more likely to gravitate towards taxis or personal driving rather than rely on time savings offered by MaaS subscription (by advance booking, seamless transfer, etc.). This inclination is particularly noticeable given the limited application of MaaS presently, leaving many individuals without an intuitive understanding of these MaaS features.

In terms of socio-demographic characteristics, all three models indicate that households possessing at least one car are more likely to subscribe to MaaS subscription. This somewhat corroborates the findings proposed by Ho et al. (2020) that frequent car users are more likely to subscribe to MaaS compared to non-car users. Individuals with a higher level of education (holding a bachelor's degree or above) and those with lower income levels also tend to show a preference for subscriptions.

The positive coefficient for income can be interpreted straightforwardly: lower-income groups may be more attracted by the potential cost savings by MaaS bundles. The result for education can be explained in multiple ways: one possibility is that individuals with a higher level of education prioritize time efficiency in travel. This is congruent with the finding of Antoniou et al. (2007), which indicates that preferences for faster modes increase with education attainment. Another plausible interpretation is that highly educated individuals are inherently more intrigued by novel services and technologies, making them more willing to experiment with emerging mobility concepts such as MaaS (Mikhaylova et al., 2021). Moreover, the results also indicate that individuals aged over 35 exhibit a higher inclination to adopt MaaS, which is likely driven by their complex work and life requirements, requiring flexible and convenient transportation solutions. Gender appears not to have a significant effect in any of the three models examined. However, the consistently negative coefficient for males implies their relatively lower tendency to embrace MaaS.

Examining mode-specific constants uncovers a clear preference shift. Initially, travelers show a strong preference for driving over other alternatives (taxi, carpool, with PT as the reference). However, this changes with MaaS introduction, leading to a greater willingness to use these alternative modes.

Table 3
Estimation results of the EUT and prospect theory-based models.

	Benchmark model		EUT-based CARA model		Prospect theory -based model	
Variable	Coef. (SE)	Sig.	Coef. (SE)	Sig.	Coef. (SE)	Sig.
MaaS bundle attributes						
Bundle price (HKD, $\times 10^{-3}$)	−7.28 (1.2)	***	−6.32 (1.3)	***	−7.19 (1.4)	***
Fare-discount for PT and carpool (%)	0.68 (0.3)	**	0.48 (0.3)	*	1.02 (0.3)	***
Fare-discount for taxi (%)	0.20 (0.2)	0.38	0.13 (0.2)	0.58	0.33 (0.2)	0.16
Time-saving for PT and carpool (%)	−4.70 (1.7)	***	−3.85 (2.4)	*	−2.91 (1.7)	*
Time-saving for taxi (%)	1.19 (0.7)	*	1.50 (0.7)	**	1.53 (0.7)	**
ASC for subscription	−0.59 (0.4)	0.14	−0.81 (0.4)	*	−2.49 (0.8)	***
$\overline{\beta}$ of deviation of ASC under uncertainty	−1.12 (0.4)	***	−0.86 (0.3)	**	−0.84 (0.3)	**
σ of deviation of ASC under uncertainty	5.37 (0.6)	***	4.73 (0.6)	***	4.10 (0.5)	***
Socio-demographic characteristics						
Age: above 35	0.58 (0.3)	**	0.66 (0.3)	**	0.70 (0.3)	**
Education: Bachelor or above	0.41 (0.3)	*	0.54 (0.3)	**	0.76 (0.3)	**
Owning at least 1 car in the household	0.68 (0.3)	**	0.66 (0.3)	**	0.79 (0.3)	***
Individual income ($\times 10^{-5}$)	−1.27 (0.5)	**	−1.43 (0.6)	**	−1.49 (0.6)	**
Male	−0.09 (0.2)	0.71	−0.19 (0.2)	0.42	−0.34 (0.3)	0.22
Mode-preference constants						
Constant for driving, PAYG	2.86 (1.7)	*	3.92 (2.2)	*	2.08 (1.6)	0.18
Constant for taxi, PAYG	0.65 (2.1)	0.75	1.88 (2.6)	0.47	−0.05 (1.7)	0.98
Constant for carpool, PAYG	0.03 (1.1)	0.98	0.29 (1.3)	0.82	1.15 (1.2)	0.33
Constant for PT, PAYG			Fixed at zero for normalization			
Constant for driving, subscription	2.13 (1.7)	0.21	3.29 (2.1)	0.12	2.14 (1.5)	0.15
Constant for taxi, subscription	5.15 (1.3)	***	7.16 (2.1)	***	4.23 (1.0)	***
Constant for carpool, subscription	2.41 (0.7)	***	3.26 (1.1)	***	2.54 (0.6)	***
Constant for PT, subscription			Fixed at zero for normalization			
$\overline{\beta}$ of deviation of driving under uncertainty	−0.29 (4.5)	0.95	0.06 (3.2)	0.98	0.14 (3.6)	0.97
σ of deviation of driving under uncertainty	17.70 (7.8)	**	14.30 (5.3)	***	12.90 (5.2)	**
$\overline{\beta}$ of deviation of taxi under uncertainty	12.40 (4.7)	***	11.40 (4.8)	**	5.65 (3.0)	*
σ of deviation of taxi under uncertainty	27.80 (7.3)	***	27.70 (8.6)	***	10.70 (4.6)	**
$\overline{\beta}$ of deviation of carpool under uncertainty	1.96 (2.2)	0.37	1.27 (2.8)	0.65	0.39 (1.4)	0.79
σ of deviation of carpool under uncertainty	10.20 (4.9)	**	10.80 (5.4)	**	4.74 (2.2)	**
Deviation of PT under uncertainty			Fixed at zero for normalization			
Non-linear transformation parameters						
$\overline{\beta}$ of travel cost ($\times 10^{-3}$)	−8.13 (1.7)	***	−10.20 (1.8)	***		
σ of travel cost ($\times 10^{-3}$)	10.10 (1.2)	***	11.50 (2.0)	***		
$\overline{\beta}$ of travel time ($\times 10^{-3}$)	−11.80 (3.4)	***	−21.50 (8.8)	**		
σ of travel time ($\times 10^{-3}$)	36.60 (3.3)	***	92.40 (24.6)	***		
r_c			0.37 (0.2)	**		
r_t			1.87 (0.6)	***		
$\overline{\beta}$ of travel cost gain ($\times 10^{-3}$)					4.98 (1.5)	***
σ of travel cost gain ($\times 10^{-3}$)					5.07 (1.3)	***
$\overline{\beta}$ of travel cost loss ($\times 10^{-3}$)					−0.17 (1.6)	0.91
σ of travel cost loss ($\times 10^{-3}$)					9.36 (4.3)	**
$\overline{\beta}$ of travel time gain ($\times 10^{-3}$)					3.00 (1.6)	*
σ of travel time gain ($\times 10^{-3}$)					4.77 (0.8)	***
$\overline{\beta}$ of travel time loss ($\times 10^{-3}$)					−4.75 (2.1)	**
σ of travel time loss ($\times 10^{-3}$)					5.17 (1.7)	***
δ_c					0.75 (0.2)	***
δ_t					0.15 (0.1)	**
Adjusted pseudo-r^2	0.242		0.251		0.335	

5.2. Variables affected by uncertainties

The estimated coefficient for the deviation of the ASC under uncertainty significantly trends negatively in all three models, suggesting that travelers exhibit increased hesitancy to subscribe to a MaaS bundle when faced with uncertain circumstances. This

hesitancy can be attributed to the heightened difficulty that respondents face in accurately assessing the potential benefits that MaaS bundles may offer them in uncertain situations. Moreover, the deviation of mode preference reveals that travelers, when confronted with uncertainty, exhibit a greater propensity to use taxis over PT. This deviation is reasonable, as taxis offer a high degree of flexibility that can aid people in navigating uncertain situations (Aarhaug and Skollerud, 2014; Rodrigues et al., 2020).

When examining the estimation of uncertain dimensions, namely travel time and cost, the coefficient signs align with our expectations. In the EUT-based CARA model, the two non-linear transformation parameters, r_c and r_t, are significantly positive, indicating a significant risk aversion towards both travel cost and time. The absolute value of r_t is more than five times that of r_c, suggesting that travelers exhibit a higher degree of risk aversion towards travel time compared to travel cost. This heightened risk aversion towards travel time could stem from the perception of time as a scarcer and less manageable resource compared to travel cost. Unforeseen delays may cause substantial disruptions to an individual's schedule, leading to heightened dissatisfaction with the travel experience. This is particularly true for commute trips, where individuals display increased sensitivity to delays and unreliability (Jarvis, 2005; Schwanen and Ettema, 2009).

Both δ_c and δ_t from the prospect theory-based model fall within the expected range of 0 and 1, representing the diminishing sensitivity to gains and losses of travel cost and travel time. The smaller value of δ_t implies that sensitivity to changes in travel time diminishes more strongly than for travel cost. This suggests that in commute scenarios, where time is a scarce resource, the priority given to saving time takes precedence over concerns about marginal cost savings. Loss aversion (λ in Eq. (8)), represented by $-\left|\frac{\beta_{(loss)}}{\beta_{(gain)}}\right|$, is 0.03 for travel cost and 1.58 for travel time, noting that the parameter for losses in travel cost is highly insignificant. Although prospect theory studies typically assumed $\lambda > 1$ to accommodate loss aversion, exceptions have been noted in previous studies (Hess et al., 2008; Latinopoulos et al., 2017). Hess et al. (2008) propose this disparity may arise from respondents' inability to evaluate individual attributes independently, as their assessments are influenced by the interplay of these attributes. Considering that travel time and cost are inherently correlated in our context, this interpretation is plausible to a certain degree. This amplified loss aversion to travel time supports our findings from the EUT-based model, where travelers exhibit a greater degree of risk aversion to travel time compared to travel cost.

The estimation of the standard deviations of random parameters sheds light on preference heterogeneity. Significant coefficients imply that the variance from the mean is statistically distinct from zero. Additionally, larger standard deviations in parameters indicate greater heterogeneity among respondents. The results reveal a spectrum of differences in how respondents perceive travel time and cost, with the scale of standard deviations for travel time indicating an exceptionally high level of heterogeneity. This diversity is also evident in the preference deviation variables for the ASC of MaaS subscription and in the varying coefficients related to mode preference.

6. Discussion

6.1. Policy implications

By providing critical insights into traveler behavior towards MaaS, this study offers the following key policy and operation strategy implications for real-world MaaS implementations:

Socio-demographic considerations: MaaS operators and policymakers should recognize the significance of demographic segmentation when shaping marketing and service strategies. For instance, given that older individuals, car owners, and those with higher education levels show a greater inclination towards subscribing to MaaS, it becomes beneficial to tailor promotional efforts and service designs specifically to appeal to these socio-demographic segments. Additionally, the interest of lower-income demographics in MaaS highlights its potential as a cost-effective mobility solution for these groups. To promote greater equity in transportation resource accessibility, policymakers should explore options such as offering subsidies or financial incentives to operators that offer cost-effective MaaS solutions to these segments.

Triggering transport mode change: The increased inclination to use alternative transport modes when MaaS bundles are accessible indicates latent demand for a variety of travel options. This suggests that the provision of integrated transportation solutions may reduce the reliance on private car usage, contributing to lower traffic congestion and carbon emissions.

Insights into risk-related behavior: Our findings concerning uncertainty and risk aversion suggest that travelers' hesitation towards subscribing to MaaS services could potentially be alleviated through the provision of clear and reliable service information. Implementing strategies such as trial periods or temporary discounts might be worthwhile for MaaS platform operators to reduce the perceived risks associated with subscribing to this innovative service. Moreover, travelers exhibit stronger risk aversion towards travel time compared to travel costs. This could be attributed to the increasing value placed on time in fast-paced societies, especially in commute trips. Therefore, MaaS operators should prioritize improving service efficiency and reliability. Government entities can also play a role by investing in infrastructure and policies that facilitate efficient MaaS operations, such as dedicated bus lanes or real-time traffic information systems.

6.2. Limitation and future work

While this study endeavors to capture parameter heterogeneity, it falls short in analyzing the heterogeneity among factors reflecting risk attitudes and diminishing sensitivity regarding travel time and cost. This limitation stems from the complexity involved in assuming these non-linear parameters adhere to a distribution in a mixed logit framework. Future research could explore alternative modeling techniques, such as latent class choice models, to more effectively investigate heterogeneous preferences towards these parameters among different socio-demographic groups. Moreover, future research should focus on refining SP survey designs or exploring alternative data collection methodologies to facilitate respondents' comprehension and evaluation of correlated attributes. Additionally, while our study primarily addresses uncertainty emanating from occasional activities (i.e., unexpected additional trips), it is worth noting that each journey itself is subject to variability in travel time and cost influenced by real-time traffic conditions. Future studies could benefit from developing more advanced survey or experimental designs and modeling frameworks to encompass a wider spectrum of uncertainties.

7. Conclusions

Uncertainties originating from occasional activities during regular commute trips make it challenging for travelers to accurately predict their upcoming travel needs. Consequently, this complexity influences the decision-making process regarding MaaS subscription and transportation mode selection. In our study, we developed models that are both EUT based and non-EUT based, incorporating linear and non-linear specifications, to estimate a spectrum of consumer behaviors under risk within the MaaS context. These behaviors include the subscription decision, changes in transport mode preferences, risk attitudes towards uncertainty, diminishing sensitivity towards gains and losses, and the presence (or lack) of loss aversion. We simultaneously considered two dimensions of uncertainty: travel time and travel cost.

The outcomes generated from all models employed demonstrate a substantial degree of consistency in terms of sign and statistical significance. Our study reveals that travelers tend to favor MaaS bundles that are either more affordable or provide substantial discounts, and they place a higher value on time savings offered by taxis rather than PT or carpool. Additionally, individuals who are older, more educated, possess private cars in their household, and have a lower individual income show a higher likelihood of subscribing to MaaS. Moreover, with the availability of MaaS bundled services, there is a noticeable shift in travelers' preferences, moving away from driving towards alternative transportation options, indicating the potential of MaaS in promoting sustainable travel patterns.

Under uncertain circumstances, travelers display heightened hesitance toward subscribing to MaaS. This apprehension can be attributed to the challenges that respondents encounter in accurately assessing the potential benefits of MaaS bundles in uncertain situations. Furthermore, in uncertain conditions, individuals show a stronger inclination towards using taxis, as taxis are perceived to provide greater flexibility that can effectively assist individuals in managing uncertain scenarios.

To conclude, our research offers valuable insights into the evolving dynamics of urban mobility, particularly in the context of MaaS. It underscores the critical role of MaaS in reshaping travel behaviors and highlights the complex interplay between consumer preferences, economic factors, and the inherent uncertainties of daily travel. These findings not only contribute to a deeper understanding of MaaS adoption patterns but also pave the way for future research and policy development aimed at fostering more sustainable and efficient urban transportation systems.

CRediT authorship contribution statement

Jianing Liu: Writing – review & editing, Writing – original draft, Data curation, Formal analysis, Investigation, Methodology. **Sisi Jian:** Writing – review & editing, Writing – original draft, Supervision, Resources, Project administration, Methodology, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Chenyang Wu:** Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Data curation, Conceptualization. **Vinayak Dixit:** Writing – review & editing, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could h8ave appeared to influence the work reported in this paper.

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Appendix A:. Screenshot of the MaaS introduction page from the survey

In this section, suppose there is a MaaS company in HK providing different packages. If you buy a package, you can get discounts on major transport modes included in the packages.

To give you a sense of how the "package" works, assuming in a normal weekday morning, Jack needs to depart from the North Point Ferry Terminal to the International Finance Centre for work. As suggested by Google Maps, Jack can take **MTR for 8 HKD with 27 min, bus for 6 HKD with 32 min, or Taxi for about 80 HKD but only 12 min.** Suppose there's a MaaS package that costs **100 HKD per week** and provides a 40% discount on public transit (Carpool, Bus, and Rail) and a 20% discount on Taxi. Moreover, this package could save Jack 10% travel time by higher priority when taking a Taxi, a special channel when taking public transit, and the privilege to use toll road if necessary. If Jack buys the package, he only needs to pay for approximately **5 HKD for MTR with 24 min, 4 HKD for bus with 29 min, or 64 HKD for taxi with 11 min**. Furthermore, **Jack can choose and pay for whatever travel mode he prefers, even a taxi, through this single APP.**

You will see 6 packages in 6 different scenarios. Different packages, with different prices, could provide you different levels of discounts. As a result, you may think of switching to a mode that you don't use very often previously, as this mode is faster and more affordable by having the discount now (e.g. Taxi). You may also consider to take a mode more often because this mode has a larger cost discount (e.g. Carpool, Bus, and Rail). In each scenario, you need to **(1)** determine the number of times you will choose to use each of the travel modes, and **(2)** determine whether you would consider to buy such a package if it were available on the market today. Your decision can be made by the tradeoff between the discounts you could get and the price of the package.

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